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Influencer Mix Strategies in Livestream Commerce: Impact on Product Sales

Xian Gu, Xiaoxi Zhang, and P.K. Kannan

Abstract

A new trend in influencer marketing is livestream commerce, which combines influencers' live video streaming with e-commerce. Little is known, though, about how firms should develop their influencer strategies to maximize product sales. Given a budget constraint, marketers may choose to employ a single, "big" influencer with a large number of followers or several small influencers. They may also employ a combination of both big and small influencers when their budget is adequate. Analyzing data from 1.3 million livestreams on TikTok, the authors find a negative interaction effect between big and small influencers when employed together due to decreased trust in big influencers and substitution effects. Further analysis reveals that livestream sales generated by a big influencer are adversely affected by small influencers who promoted the same product previously, but not the other way around. In addition, big influencers can reach a much larger audience, whereas small influencers are more effective at increasing the audience's conversion rates. Big and small influencers are also different in terms of how product and campaign characteristics moderate their sales effectiveness. Finally, the scenario analysis explores how the profitability of influencer strategies hinges on the influencer mix and their costs. This research is one of the first to investigate the sales effectiveness of influencer marketing and to provide general guidelines for developing influencer mix strategies.

Keywords

livestreaming, influencer marketing, social commerce, social shopping, user-generated content, social media, product sales

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The rapid development of high-speed internet connectivity and the growth of online influencers have introduced a popular and impactful form of commerce, "livestream commerce." In a typical livestream, an influencer demonstrates and promotes products while providing entertainment content such as singing and talk shows. Viewers can interact with the influencer and other viewers in real time by asking questions, leaving comments, and even giving virtual gifts or tips to the influencer. Such livestreams often feature exciting music and attractive price discounts, with influencers encouraging viewers to make immediate purchases through onscreen links using phrases such as "buy it now." It is not surprising, therefore, that livestream commerce has grown at a phenomenal rate over the last few years, with multiple social media platforms and major retailers adopting it. Facebook, TikTok, and YouTube have introduced their respective livestream shopping services on their platforms. Amazon launched Amazon Live in early 2019 and features livestreams on its home page, competing with TikTok in social shopping (Bobrowsky, Herrara, and Wells 2023). Walmart has, since 2020, collaborated with TikTok to stage multiple livestream shopping events. Many other retailers and firms (e.g., Nordstrom, Macy's, LG, and Petco) have also embraced livestream commerce (Larson 2021).

Livestream commerce differs from traditional e-commerce in the sense that visitors are directed by the influencer to the firm's website for real-time purchases while the livestream is still ongoing. The livestreams typically last several hours, with influencers promoting multiple products, one after another in a single session. The resulting sales from these livestreams can be viewed as promotional sales, which depend on the influencers' reach, reputation, and persuasiveness in driving purchases. Unlike other forms of influencer marketing, it is well known to consumers that livestream commerce is often sponsored by firms. Therefore, our research focuses on sponsored livestreams in which influencers explicitly promote products

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to viewers and viewers understand that livestreams are not organic or spontaneous.

Despite the popularity of livestream commerce, there are still unanswered questions about the effectiveness and design of livestream influencer campaigns. Influencers' impact varies greatly in terms of the number of followers each attracts, which largely determines their market value. Some firms use exclusively big influencers with a large number of followers, some solely use small influencers with fewer followers, and others use a combination of both. Since the market value and fees of influencers can vary significantly[1] and the marketing budget is usually a constraint, the decision to go with a specific strategy is complex (Schmidt 2018). Specifically, a key question managers grapple with is whether they should employ a single big influencer or several small influencers (Goldenberg et al. 2022).

A big influencer may achieve broad reach for a livestream, and if the influencer can direct a significant proportion of the viewers to purchase the promoted product, collaborating with one big influencer can be an effective strategy. However, while the broad reach of big influencers in raising awareness has been shown in existing literature (Hinz et al. 2011), there are mixed findings for the engagement they can elicit (Wies, Bleier, and Edeling 2023), and there has been only scant research on the conversions they can drive among viewers. In contrast, collaborating with multiple small influencers can be an attractive strategy because they charge lower fees and may convert a large proportion of their followers given their persuasive powers and close relationships with followers (Haenlein et al. 2020). Both awareness and conversions are important for livestream commerce, where the entire customer journey —from awareness to consideration to the actual purchase— can unfold in a single session, thus meriting focused research.

Moreover, when an adequate budget is available, an important question is whether it is better to use a combined strategy of big and small influencers rather than one or the other exclusively. One may argue that the combined strategy should yield positive synergistic effects by leveraging their unique strengths at different stages of the customer purchase journey. Whereas big influencers can create awareness, small influencers are effective in closing sales. Consequently, this combined strategy has the potential to generate more conversions than using either big or small influencers exclusively. However, it could also be argued that the interaction effects between big and small influencers are negative due to consumers' decreased trust in influencers and substitution effects, especially when consumers learn that multiple influencers are promoting the same product.

What is the relative sales effectiveness of big and small influencers in livestreaming? Should marketing managers employ one type of influencers exclusively or a combination of both types? How should firms change their influencer strategies in the presence of price discounts and copromoted products and categories? What mechanisms may be driving the answers to the preceding questions? We examine these research questions systematically by obtaining a unique and rich data set from TikTok in China (Douyin), a social media app for livestreaming and short video sharing. We gather detailed information about products, influencers, and livestreams on TikTok during a five-month period from January 28 to June 28, 2021. The data contain many instances of diverse influencer strategies (big-influencer-only, small-influencer-only, and combined strategies), each of which was adopted by a significant portion of firms. We model product sales through livestreams, each of which has its unique purchase link, allowing us to link product sales directly to the influencer and the livestream.

Several important findings emerge from our research. First, big influencers are able to attract a much larger audience for livestreams, whereas small influencers are more effective at increasing the audience's conversion rates. However, a big influencer can generate significantly more sales than a small influencer as the big influencer's larger audience reach more than compensates for the lower conversion rate. Second, when a combined strategy is adopted for livestreams, there is a negative interaction effect between big and small influencers. This suggests decreased consumer trust in big influencers, which we confirm using an experiment, and substitution effects between influencers. Further analysis also reveals that livestream sales generated by a big influencer are adversely affected by small influencers who promoted the same product previously, but not the other way around. In addition, we highlight differences between big and small influencers in terms of how product and campaign characteristics moderate their sales effectiveness. Notably, compared with small influencers, big influencers benefit more from promoting a limited number of products, focusing on a single product category, and offering deep discounts in livestreams. Finally, we collect the hiring costs of influencers and conduct a profit-budget scenario analysis. In general, the combined strategy and the big-influencer-only strategy perform similarly and beat the small-influencer-only strategy. However, when the costs of hiring big influencers are high and the costs of hiring small influencers are low, the combined strategy outperforms the big-influencer-only strategy but gets dominated by the small-influencer-only strategy.

Our findings are novel as this study is one of the first to empirically assess the sales effectiveness of influencer marketing, especially the influencer mix. We emphasize that employing both big and small influencers to promote a product may not necessarily be an effective strategy to maximize firms' return on investment, which may contradict the early beliefs of many marketers (Haenlein et al. 2020; Schmidt 2018). Also, we empirically show the relative strengths of big and small influencers in attracting audiences and driving conversions and highlight their differences based on various moderating effects.

This article contributes to the literature on livestreaming, which is still in its infancy. The majority of studies in this field have focused on understanding influencers' and viewers' behavior in livestreams (e.g., Lin, Yao, and Chen 2021; Lu

[1] On Instagram, for example, influencers with more than one million followers could charge over $10,000 per post, whereas influencers with fewer than 50,000 followers are generally compensated $10–$500 per post (Geyser 2023).

Table 1. Comparisons Between Prior Studies and Our Study.

Research	Context	Sales Impact	Influencer Marketing	Influencer Mix
Karagür et al. (2022)	Instagram post	×	✓	×
Hughes, Swaminathan, and Brooks (2019)	Blog	×	✓	×
Lin, Yao, and Chen (2021)	Livestream	×	×	×
Liu (2023)	Livestream	✓	×	×
Lu et al. (2021)	Livestream	×	×	×
Wies, Bleier, and Edeling (2023)	Instagram post	×	✓	×
Our study	Livestream	✓	✓	✓

et al. 2021). Recently, Liu (2023) investigates dynamic coupon targeting in the context of livestream commerce but does not study the role of influencers. To the best of our knowledge, our study is among the first to investigate the sales effectiveness of influencers in livestream commerce and thus provides important implications for researchers as well as practitioners.

Our article also adds interesting insights to a growing stream of research on influencer marketing (Hughes, Swaminathan, and Brooks 2019; Karagür et al. 2022; Lee and Junqué de Fortuny 2022; Leung, Gu, Li, et al. 2022, Zhang, Chintagunta, and Kalwani 2021). Specifically, we provide empirical support for hypotheses in extant research that big influencers are more effective at reaching large audiences than in closing sales, which small influencers are more effective at. Our research also sheds light on how influencers can improve their sales effectiveness by carefully designing their campaigns. Moreover, our research is distinct from previous studies in an important way: we measure outcomes by using product sales that are linked directly to each influencer, as opposed to using engagement or aggregate sales (Wang et al. 2023a). A few recent studies link influencers to product sales (Beichert et al. 2022; Wang et al. 2023b), but they do not examine in great detail the influencer mix, which is the focus of our research. We provide a comparative review of previous studies on livestreaming and influencer marketing in Table 1, highlighting our contributions.

Research Background and Theoretical Framework

Elements of Livestream Commerce

Livestream commerce creates an interactive and engaging shopping experience for consumers by combining live video streaming and online shopping with the power of social media influencers. Livestream commerce allows viewers to ask questions, receive immediate product information, see demonstrations, and interact with the influencer or other viewers, fostering a sense of community and urgency. These dynamic and immersive experiences and the bonds between influencers and followers distinguish livestream commerce from other shopping formats and infuse traditional e-commerce with the personal touch of influencers and real-time interaction that it lacks. Livestream commerce serves as a unique platform for firms and influencers to connect with their audience, build trust, and drive sales in a more captivating and entertaining manner compared with standard e-commerce platforms.

The role of influencers in livestream commerce shares common ground with other contexts of influencer marketing, which have evolved from a growing skepticism about traditional forms of advertising (Karagür et al. 2022). Even though followers may never meet an influencer in person, influencers share insights into their personal lives, provide advice, and share funny stories with their followers very much as real friends would do, characterizing a parasocial relationship (Audrezet, De Kerviler, and Moulard 2020; Sokolova and Kefi 2020). Therefore, influencers tend to appeal to followers in a more relatable or down-to-earth manner as opposed to what occurs with aloof celebrities such as athletes and actors (McQuarrie, Miller, and Phillips 2013). Thus, influencers intertwine product endorsements into their personal narratives, making their content more authentic and reflective of their personal tastes (Leung, Gu, and Palmatier 2022).

Credibility is one of the most critical qualities of influencers, which depends on an influencer's perceived trustworthiness and expertise (Sokolova and Kefi 2020; Zhang, Chintagunta, and Kalwani 2021). Because influencers typically start as ordinary social media users with no status, their content appears less commercial and more trustworthy than content generated by marketers and firms. Followers value influencers' seemingly intrinsic and noncommercial motivations (Leung, Gu, Li, et al. 2022). Based on their trust in influencers, consumers actively seek recommendations from the influencers they follow and make purchases based on those recommendations (Traackr 2022). The perceived expertise reflects followers' perceptions of the influencer's qualifications to discuss a given subject and offer advice, which depends heavily on the content the influencer has consistently created (Hughes, Swaminathan, and Brooks 2019; Lou and Yuan 2019). Belanche et al. (2021) provide an excellent example of an influencer promoting a product that is inconsistent with her social image built on her past social media posts, resulting in criticisms of both the influencer and the brand.

Attractiveness is another essential influencer quality (Lou and Yuan 2019), which depends on two components: whether an influencer is seen as personable and appealing (reflecting likability) and whether followers perceive the influencer as similar to them (reflecting perceived similarity) (Leung, Gu, and Palmatier 2022; Percy and Rosenbaum-Elliott 2016). Attractiveness is particularly important in influencer

marketing because it contributes to emotional advertising appeals and has a significantly positive impact on brand attitude (Bakker 2018).

Despite the similarities, livestream commerce also has clear differences in influencer roles when compared with other contexts of influencer marketing. First, consumers are well aware that influencers are sponsored by firms to promote products in livestreams, which are often designed to be shopping-oriented. Thus, we would not expect the commercial nature of livestream commerce to undermine the credibility of influencers. Rather, consumers, especially followers, would trust an influencer's knowledge, high standards, and good taste in product selection, and thus perceive the influencer as credible in the shopping context. Second, influencers in other contexts focus on engaging consumers and raising product awareness as part of consumers' path to purchase, whereas the entire customer journey can unfold within a single livestream, leading to immediate sales. Thus, the effectiveness of influencers in livestream commerce is predominantly measured by the actual product sales, rather than intermediate metrics such as engagement. Third, influencers usually promote multiple products in a single livestream. Thus, the number of products promoted together, the categories copromoted, and the discounted prices influencers offer to viewers all impact the sales outcomes. These distinct characteristics of influencers in the context of livestream commerce make it unique and worthy of research in its own respect.

Sales Effectiveness of Big and Small Influencers

In the livestreaming context, we argue that the sales performance of influencers depends on their ability to reach a large number of prospective consumers as well as drive conversions. Extant literature has shown that big influencers with many followers are effective at boosting information diffusion (Hinz et al. 2011). Furthermore, livestreaming underscores the importance of entertaining followers and viewers (Hilvert-Bruce et al. 2018), and thus places influencers at the center of the shopping process in so-called "retailtainment" (Schwarz 2020). Compared with small influencers, big influencers have established their wider appeal by acquiring a larger follower base via their personality, talent, entertaining interactions, and so forth. Therefore, we anticipate that big influencers will be more effective than small influencers at reaching and entertaining a large number of prospective consumers.

Conversely, followers are more likely to retain strong and close bonds with small influencers who appear more approachable and more similar to themselves than big influencers who have a wider, more diverse audience (Leung, Gu, and Palmatier 2022). For example, small influencers can devote more time to individual interactions, and thus they can foster more intimate relationships with their followers (De Veirman, Cauberghe, and Hudders 2017). Previous studies present evidence that viewers enjoy taking part in intimate conversations with influencers (Lu et al. 2021), and a close-knit relationship with followers can generate persuasive power to close the deal (Haenlein et al. 2020). Consequently, we expect that small influencers generate higher conversion rates than big influencers given their intimate relationships with followers.

Table 2. Average Audience Reach and Conversion Rate of Big and Small Influencers.

Measures	Big Influencers	Small Influencers	Ratio
Audience reach (viewers)	276,998	61,491	4.50
Conversion rate (live sales/viewers)	.263%	.549%	.48

As an influencer demonstrates and promotes a product, a viewer considers information in real-time interactions with the influencer and other viewers, with the process ultimately culminating in a purchase, all in one livestream setting. Despite having lower conversion rates than small influencers, big influencers have a much greater audience reach (see Table 2). Therefore, we anticipate that big influencers' much larger reach will more than compensate for their lower conversion rate, enabling them to generate more sales. Additionally, impulse purchases are common in livestream commerce, which can also lead to additional conversions when big influencers attract large audiences (Lo et al. 2022).

Interaction Effects Between Big and Small Influencers

It is also common for firms to employ both big and small influencers for the same product (Schmidt 2018).[2] Deploying such a combined strategy could, however, increase the likelihood that consumers notice multiple influencers promoting the same product due to follower overlap and the platform's recommendation algorithm (Goldenberg et al. 2022). If a firm only employs small influencers, there may be no significant audience overlap because their follower bases are small. Big influencers can, however, attract more than one million followers, and thus their audiences are more likely to overlap with those of other influencers when the combined strategy is employed. Furthermore, platform algorithms strive to understand what products a consumer may be interested in and will strategically recommend relevant livestreams to consumers (Zhao 2021). As a result, even if follower overlap is minimal, consumers may still learn from the platform's recommendations that multiple influencers are promoting the same product. However, consumers' awareness of this fact can have either a positive or negative impact on sales, depending on how big and small influencers wield their influence in consumers' purchase funnels.

On the one hand, hiring both big and small influencers could generate positive interaction between the two, resulting in an overall synergistic effect. Hiring a big influencer may enable a firm to attract a larger audience and thus increase awareness of the firm's products and orient consumers positively to a

[2] For example, 16% of the products in our data were promoted by both big and small influencers.

purchase decision even if the influencer is not able to convert them in that livestream. At the same time, small influencers can be critical when firms seek to reach consumers in niche markets, and they often generate higher conversion rates (Haenlein et al. 2020). Given that big and small influencers each have their own strengths and effectiveness at different stages of the purchase funnel, it may become easier for a small influencer to persuade consumers who have become aware of a product from a big influencer to purchase the product. As a result of the complementary effort along the purchase funnel, utilizing both types of influencers may result in higher sales than just relying on either big or small influencers exclusively, especially when the consumers reached by the big and small influencers overlap.

On the other hand, when consumers notice multiple influencers promoting the same product, they may perceive the promotion as mere advertising and question the influencers' genuine motivations for selecting the product (Bakker 2018). We anticipate that big influencers who heavily rely on followers' trust and carefully curated images would be more susceptible to this phenomenon, compared with small influencers who have more time to interact with their smaller follower base and foster closer bonds. As a result, we expect diminished trust in big influencers to negatively affect sales performance when both big and small influencers are employed. In addition, if big and small influencers wield similar types of impact on prospective consumers, hiring both types of influencers could generate substitution effects. That is, a portion of one influencer's audience could purchase from other influencers who promote the same product. The substitution effects can be amplified when there is a high overlap in followers, which is more likely when big influencers are involved. Consequently, a negative interaction effect between big and small influencers can arise. Finally, utilizing a diverse group of both big and small influencers may undermine the overall fit between the product and influencers, thereby diminishing the effectiveness of livestreaming.

Our discussions suggest that the impact of employing both big and small influencers for promoting the same product could go either way: positive or negative. We expect our empirical analysis to provide deeper insights into this.

Data

Institutional Setting

We collect data for our study from livestream commerce on TikTok in China (Douyin), which was first launched in September 2016 and had reached more than 600 million daily active users by 2021. To monetize its high traffic, TikTok entered the e-commerce industry in early 2018, allowing firms to sell products through livestreams and short videos. Later, TikTok introduced its own online stores, and e-commerce transactions on TikTok reached $77 billion in gross merchandise value in 2020, making it a leading social media and e-commerce platform worthy of research (Cheng 2021).

A typical livestream on TikTok is presented in a virtual showroom set up by an influencer. TikTok influencers often announce their upcoming livestreams a few days ahead of time, including the products or categories they plan to pitch. Users may join the showroom while the livestream is ongoing, and influencers can provide entertainment content such as singing, dancing, and talk shows to continuously fascinate and entertain their audiences (Lin, Yao, and Chen 2021).

Figure 1, Panel A, displays a screenshot of an influencer promoting a smartwatch to her viewers in a TikTok livestream. It is common for viewers to ask questions about products in live comments and receive immediate answers from the influencer. Figure 1, Panel B, shows a list of products promoted in the livestream, including a smartwatch, a floor scrubber, children's clothes, and women's shorts. In a livestream, an influencer can promote multiple products, which may fall into the same or different categories and be sold by the same or different firms. In our research context, retailers (i.e., online TikTok stores) are referred to as firms because they make decisions about influencer strategies. Viewers can see each product's discounted price and buy it immediately by clicking through a purchase link. The discounted prices are offered only during the livestream to viewers, who must make purchases through the livestream to avail themselves of the discounts. Consumers who purchase directly from a TikTok store without clicking through a livestream pay regular prices. When viewers click on a listed product, the product information page appears, as shown in Figure 1, Panel C. By visiting this page, viewers can learn more about the product and the firm.

TikTok also established a marketing platform, Xingtu (https://www.xingtu.cn), to connect firms and TikTok influencers. On this platform, firms can access all TikTok influencers, manage their contracts with influencers, review marketing and sales data related to their influencer strategies, and make payments to influencers. To assist firms in selecting suitable influencers, Xingtu displays detailed information about each influencer, such as follower size, historical sales performance, viewership, promoted product categories, and hiring costs. A typical way for firms and TikTok influencers to collaborate is that a firm searches for influencers on Xingtu based on its marketing goals and budget, as well as influencer information. The firm can contact influencers on Xingtu and apply to be promoted in the influencers' planned livestreams, which may have specific themes (e.g., Mother's Day gifts). In most cases, the influencer and the influencer's operations team decide on the schedule of their livestreams (e.g., time and duration) as well as what and how many products to promote in a livestream. Firms often do not know in advance what other products will be promoted in the same livestream. While this way of collaboration can be used to hire any big or small influencer, big influencers usually have more bargaining power than small influencers when it comes to negotiating with firms for the hiring costs (e.g., the fixed cost and commission) and the price discounts of products. However, firms could offer a take-it-or-leave-it contract specifying a fixed cost, a commission rate, and price discounts to small influencers.

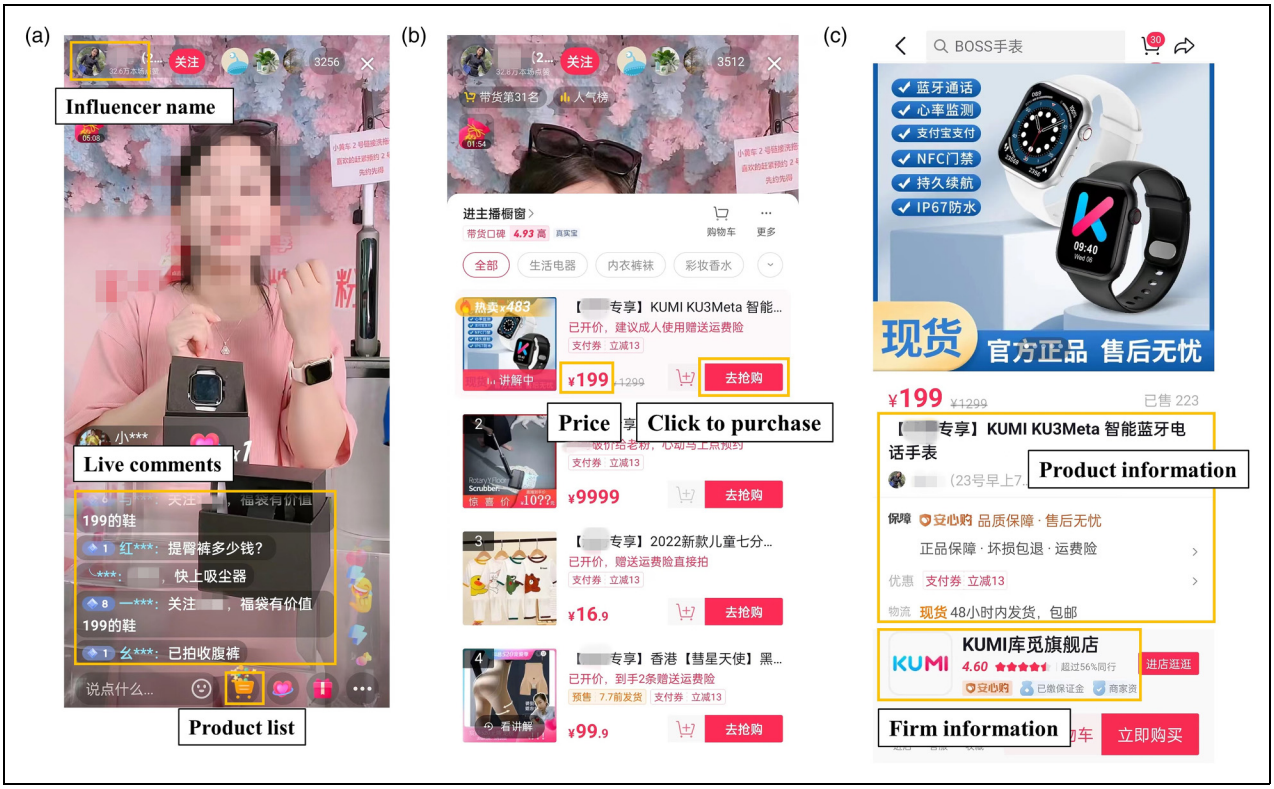


Figure 1. An Example of Livestream Commerce on TikTok.

Data Sample

We obtain a unique and rich data set on 108,001 products live-streamed on TikTok between January 28 and June 28, 2021.[3] This data set covers all products that were among the top 2,000 daily best-sellers on TikTok for at least one day over the course of our data period. The observed products cover 18 product categories. As summarized in Table 3, the "clothing, shoes, bags, and accessories" category accounts for 42.02% of products in our data, followed by the "beauty and personal care" and "grocery and gourmet food" categories.

We gather detailed information about all livestreams and influencers related to these products. We transform our data into a panel at the product and date levels, generating a total of 1,849,619 observations. The daily panel enables us to include multiple influencers within each observation, allowing us to not only compare the performance of big and small influencers but also examine the interaction effect between them. The panel structure is unbalanced given that the products were promoted in livestreams over a varying number of days. As there is no observation attrition or rotation, the estimates from our model with fixed effects are consistent when applied to this unbalanced panel.

Variables

Live sales. Our dependent variable is each product's daily unit sales during livestreams. In contrast to previous studies that rely on user engagement to measure the outcomes of influencer strategies, we observe sales directly linked to each product, influencer, and livestream. Additionally, during our data period, all product purchases in TikTok livestreams were made exclusively from TikTok stores as TikTok does not allow livestream influencers to use purchase links from other platforms.

Influencer strategies. Firms often debate whether to collaborate with a single big influencer or multiple small influencers (Goldenberg et al. 2022). As a result, we use a dummy variable, BigInf_{it}, to capture whether product i was livestreamed by any big influencer(s) on day t. NumSmallInf_{it} is the number of small influencers that livestreamed product i on day t. We follow the industry definition and define big influencers as those who have more than one million followers (Ehlers 2021). Based on this definition, 5.2% of the influencers in our data are big influencers, accounting for 33.8% of our observations. In robustness checks, we use various thresholds to define big influencers and find consistent results.

Two alternative sets of variables can also be used to describe influencer strategies. First, we can replace the dummy variable, BigInf_{it}, with the number of big influencers (NumBigInf_{it}) to capture the effects of big influencers. However, the correlation between NumBigInf_{it} and NumSmallInf_{it} is high (.64). Also, it is reasonable to use BigInf_{it} given that 93% of the observations have no or only one big influencer. The second alternative set of variables consists of the number of influencers (NumInf_{it}) and their average number of followers (Follower_{it}). However,

[3] The data set was provided by YouCloud Technology Limited.

Table 3. Product Categories.

No.	Product Category	Number of Products	Percentage of Products
1	Clothing, shoes, bags, and accessories	45,377	42.02%
2	Beauty and personal care	16,652	15.42%
3	Grocery and gourmet food	14,721	13.63%
4	Household essentials	10,212	9.46%
5	Baby, kids, and maternity	6,205	5.75%
6	Watches and jewelry	3,905	3.62%
7	Home improvement	3,185	2.95%
8	Culture and entertainment	2,395	2.22%
9	Electronics and appliances	2,252	2.09%
10	Sports and outdoors	1,498	1.39%
11	Flowers and gardening	707	.65%
12	Automotive parts and accessories	371	.34%
13	Health and medicine	288	.27%
14	Education and training	132	.12%
15	Pets	43	.04%
16	Fine art and collectibles	25	.02%
17	Domestic services	23	.02%
18	Traveling and accommodations	10	.01%

their implications are less intuitive. A firm is unlikely, for example, to consider what influencer(s) it should hire to increase the average number of followers of its influencers by a particular level. Therefore, we use these two alternative sets of influencer variables in robustness checks rather than in the main analysis.

Campaign characteristics. To control for competing products in the same livestream campaign, we include the number of products promoted in a livestream ($NumPdt_{it}$) and the category focus of a livestream with respect to product i ($CatFocus_{it}$). While a larger number of promoted products usually means that product i faces more competitors and receives less attention from consumers, it may also attract higher traffic to the livestream. $CatFocus_{it}$ is measured as the fraction of products promoted in the livestream that fall into the same category as product i. The higher the fraction is, the higher the livestream's category focus is with product i, and the more competitors product i may face. We also account for the depth of price discounts ($Discount_{it}$) for focal products in livestreams. In addition, we control for the duration ($Duration_{it}$) and time of day of a livestream ($Time_{it}$). We average the campaign variables across multiple livestreams that promote product i on day t.

Other control variables. Firms can also hire influencers to post short videos on TikTok to promote their products. In short videos, users do not interact with the influencer and other users or make purchases in real time. Given that every product in our data was promoted by at least one short video, we must control for the effects of promotional short videos. Specifically, we control for the total number of short videos posted on day t to promote product i ($NumVideo_{it}$) and the average number of likes received by these short videos ($VideoLike_{it}$). The number of short videos reflects a product's marketing intensity on TikTok, and the number of likes measures the popularity and level of engagement of those short videos.

In Table 4 we report the definitions of and summary statistics for all variables, constructed at the product and date levels. In the model estimation, we divide $NumPdt_{it}$ and $VideoLike_{it}$ by multiples of 10 for scaling purposes. Moreover, we use multiple time windows (1, 7, and 14 days) in our main analysis, taking into account two important aspects. First, firms may spread out their influencer strategies over several days. Second, influencer strategies may have temporal and spillover effects on sales. Specifically, we define all variables based on livestreams between day t and day $t-k$, with $k = 0, 6, 13$.

Data Patterns

We highlight several interesting data patterns that motivate our empirical analysis. First, we display the distribution of products with varying livestream frequencies in Figure 2. Over the five-month data period, more than half of the products had fewer than 10 livestreams, a considerable number of products had between 10 and 100 livestreams, and only a small portion of products had more than 100 livestreams.

Second, over the five-month data period, 29% of the observed products involved exclusive collaboration with big influencers and were livestreamed by between 1 and 13 big influencers throughout the period. Meanwhile, 55% of the products were promoted exclusively by small influencers and were promoted by between 1 and 719 small influencers throughout the period. In addition, 16% of the products were livestreamed by both big and small influencers, with an average of 5 big influencers and 33 small influencers. The preceding data pattern reveals that both big and small influencers are commonly hired by firms. Moreover, a considerable number of firms use a combination of the two strategies. Table 5 shows the number of products broken down by the number of big and small influencers who promoted them. This information suggests that it is crucial to understand the effectiveness of various strategies separately as well as in combination.

Furthermore, we compare the audience reach and conversion rate between big and small influencers. Table 2 presents the average values of the number of viewers (i.e., audience reach) and the ratio of live sales to viewers (i.e., conversion rate) of big and small influencers in a livestream. In line with our theoretical arguments, we find that big influencers can reach about five times the audience of small influencers on average. Small influencers, in contrast, have an average conversion rate that is double that of big influencers. The two comparisons are also consistent with our argument that a big influencer's larger audience reach may more than compensate for their lower conversion rate.

Finally, Table 6 shows that the product categories are similar for products promoted by different influencer strategies. However, products promoted exclusively by big influencers have the highest daily live sales. Products promoted exclusively by

Table 4. Variable Definitions and Summary Statistics.

Variable	Definition	Mean	SD
Dependent Variable			
$LiveSale_{it}$	Product i's unit sales during livestreams on day t	460	1,653
Influencer Strategies			
$NumInf_{it}$	Number of influencers who livestreamed product i on day t (i.e., focal influencers)	3.0	9.0
$NumSmallInf_{it}$	Number of small focal influencers (<1 million followers)	2.4	8.0
$NumBigInf_{it}$	Number of big focal influencers (≥1 million followers)	.5	1.5
$BigInf_{it}$	A dummy variable that equals 1 if any focal influencer is big	.3	.5
$Follower_{it}$	Average number of followers of focal influencers	1,206,485	3,138,049
Campaign Characteristics			
$NumPdt_{it}$	Number of promoted products in livestreams that promoted product i on day t (i.e., focal livestreams)	52.3	35.1
$CatFocus_{it}$	Fraction of products that fell into the same category as product i out of all products promoted in focal livestreams	.8	.3
$Discount_{it}$	Depth of price discounts for product i in focal livestreams	.38	.35
$Duration_{it}$	Duration (in hours) of focal livestreams	5.4	5.2
$Time_{1,it}$	A dummy variable that equals 1 if any focal livestream was hosted between 12 a.m. and 6 a.m.	.6	.5
$Time_{2,it}$	A dummy variable that equals 1 if any focal livestream was hosted between 6 a.m. and 12 p.m.	.6	.5
$Time_{3,it}$	A dummy variable that equals 1 if any focal livestream was hosted between 12 p.m. and 6 p.m.	.8	.4
$Time_{4,it}$	A dummy variable that equals 1 if any focal livestream was hosted between 6 p.m. and 12 a.m.	.8	.4
Other Control Variables			
$NumVideo_{it}$	Number of short videos posted on day t to promote product i	.4	2.2
$VideoLike_{it}$	Average number of likes received by the promotional short videos	279	6,353
Number of products		108,001	
Number of observations		1,849,619	
Number of influencers		80,274	
Number of livestreams		1,350,403	

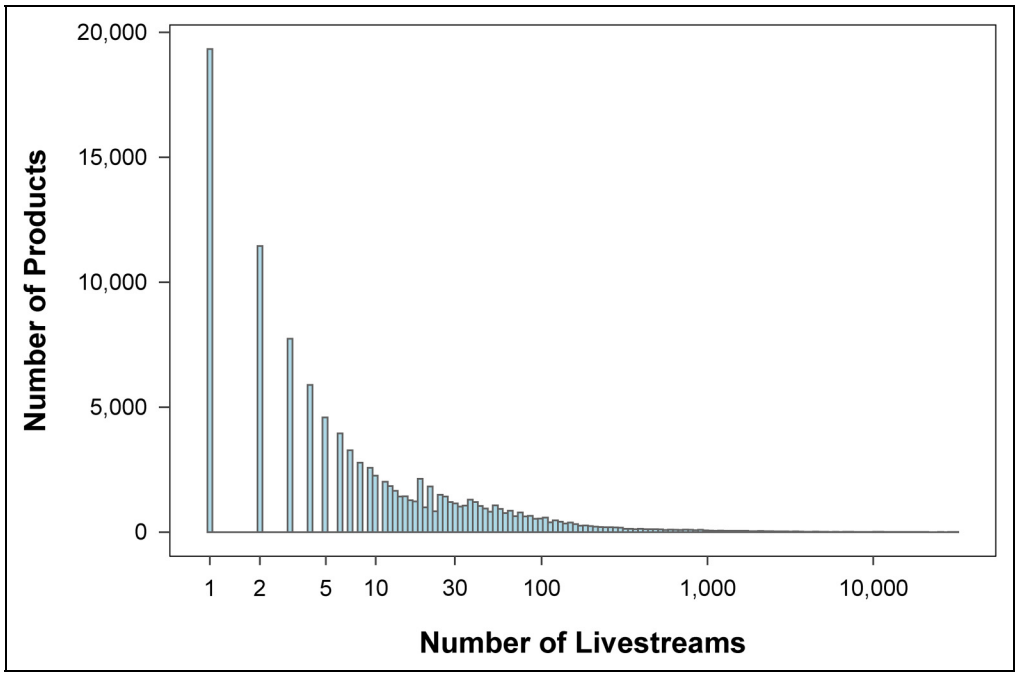


Figure 2. Distribution of Products by Livestream Frequency.

Table 5. Distribution of Products by Influencer Mix.

	Influencer Strategies		
Number of Influencers	**Big Influencers Only**	**Small Influencers Only**	**Both Types**
One big influencer	29,284	—	8,614
Two to four big influencers	2,017	—	5,112
Five or more big influencers	59	—	3,654
One small influencer	—	46,775	5,605
Two to four small influencers	—	10,224	4,099
Five or more small influencers	—	2,262	7,676
Total	31,360 (29%)	59,261 (55%)	17,380 (16%)

Table 6. Comparing Key Variables Across Influencer Strategies.

		Mean Values			
Influencer Strategies	**Top Categories**	**LiveSale**	**NumPdt**	**CatFocus**	**Discount**
Big influencers only	Clothing, beauty, food, household essentials	1,426	50	67%	43%
Small influencers only	Clothing, beauty, household essentials, food	771	44	84%	47%
Both types	Beauty, household essentials, food, clothing	628	51	64%	32%

small influencers have the smallest number of promoted products, the highest category focus, and the largest price discounts in a livestream. Products promoted by both big and small influencers generate the lowest daily live sales, the most promoted products, the lowest category focus, and the least attractive price discounts. However, the observed variations could be related to the fact that firms and products adopting a particular influencer strategy are different from those using other strategies. As a result, we must account for unobserved product- and firm-related heterogeneity when we estimate the effects of influencer strategies.

Empirical Approach

Model Specification

We consider a fixed-effects panel model to estimate how firms' influencer strategies are associated with their product sales in livestream commerce. We take advantage of the panel structure of our data to account for any time-invariant factors related to the products and firms (e.g., product quality and firm reputation) that may impact product sales. Specifically, our dependent variable, $\log y_{it}$, is the log of product i's unit sales during livestreams on day t.

$$\begin{aligned}\log y_{it} = {} & \alpha + \beta_1 \times \text{BigInf}_{it} + \beta_2 \times \text{NumSmallInf}_{it} \\ & + \beta_3 \times (\text{BigInf} \times \text{NumSmallInf})_{it} + \gamma x_{it} \\ & + \mu_i + \lambda_t + \epsilon_{it}.\end{aligned} \tag{1}$$

The variables of interest are BigInf_{it}, NumSmallInf_{it}, and their interaction term, jointly capturing the effects of the various influencer strategies. To better interpret the coefficients, we write out the partial derivatives of the regression equation with respect to BigInf_{it} and NumSmallInf_{it}:

$$\frac{\partial \log y_{it}}{\partial \text{BigInf}_{it}} = \beta_1 + \beta_3 \times \text{NumSmallInf}_{it}, \tag{2}$$

$$\frac{\partial \log y_{it}}{\partial \text{NumSmallInf}_{it}} = \beta_2 + \beta_3 \times \text{BigInf}_{it}, \tag{3}$$

where β_1 captures the average effect of employing any big influencer(s) on product live sales when no small influencer is employed (big-influencer-only strategy), and β_2 captures the average effect of employing one additional small influencer when no big influencer is employed (small-influencer-only strategy). The coefficient β_3 reflects the incremental changes in the sales effectiveness of big and small influencers when a combined strategy is adopted. We are especially interested in examining these incremental effects because they can be either positive or negative based on our theoretical considerations.

We control for a variety of campaign and product characteristics in x_{it} that may also impact live sales, such as the number of products promoted in livestreams, the category focus of livestreams, and price discounts. We use product-specific fixed effects, μ_i, to account for any time-invariant product characteristics (e.g., product category, regular price, and quality) and firm-related factors (e.g., firm reputation and marketing resources). In addition, our model incorporates multiple time dummy variables in λ_t for Chinese national holidays, days of the week, and month fixed effects to control for seasonality and time trends. These time dummy variables capture the unobserved factors in consumer demand shifts that are impacted by time. For example, one would expect consumers to have more time on holidays to watch livestreams and thereby be more likely to make purchases on those days. Finally, ϵ_{it} is an

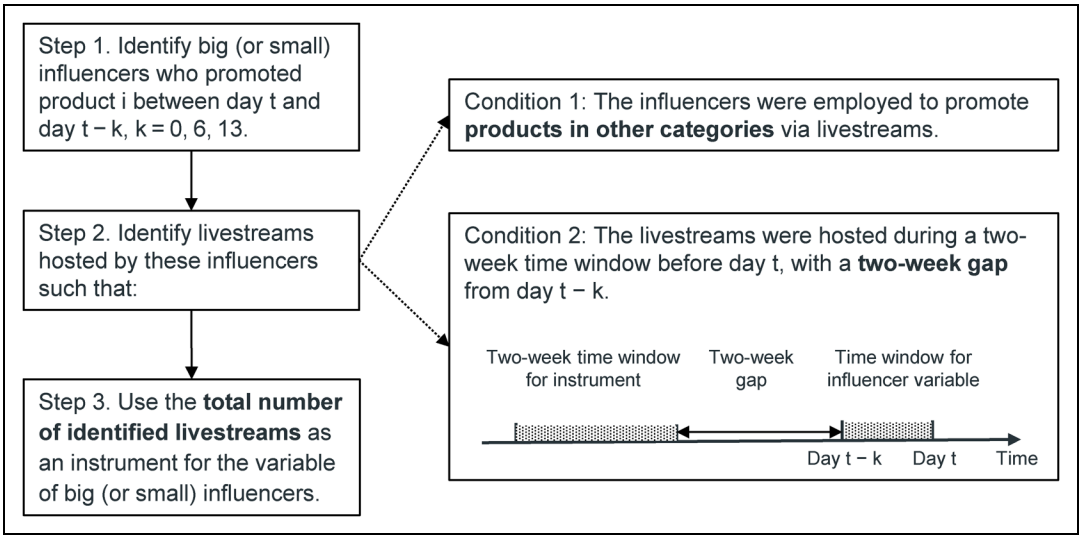


Figure 3. Construction of Instrumental Variables.

idiosyncratic error term, which is assumed to be correlated across periods for the same product and independent across products.

Table W1 in the Web Appendix displays a correlation matrix of our variables. We find that the correlations between live sales and influencer variables are significant ($p < .01$). The average variance inflation factor of our model is 3.12, which indicates no threat of multicollinearity in the model.

Endogeneity

Several types of unobserved factors might influence both live product sales and the influencer variables, $BigInf_{it}$ and $NumSmallInf_{it}$. First, unobserved product characteristics such as quality, functionality, and brand awareness may affect product sales and influencer strategies. For example, high-quality products typically sell in greater volume than low-quality products. Also, big influencers are more willing to promote high-quality products as they have stronger incentives to maintain their good reputations than small influencers, who face fewer choices regarding the products they might promote. Second, unobserved firm-level factors, such as their reputations and scale, could also affect product sales and influencer selection. Some consumers, for instance, may prefer buying products from large firms. It is often expensive to hire big influencers, and therefore large firms with large marketing budgets are more likely to be able to afford them. Finally, it is also possible that firms coordinate influencer strategies with other marketing activities such as advertising, which could drive product sales but are unobserved in our data.

We leverage two approaches to address the endogeneity concerns. First, we rely on product-specific fixed effects, μ_i, to control for any unobserved time-invariant product- and firm-level heterogeneity. Following standard microeconomic applications (Wooldridge 2010), we allow arbitrary correlations between μ_i and the explanatory variables. The fixed-effects estimation uses the within transformation to cancel out μ_i.

Second, we use instrumental variables to address the remaining endogeneity issues associated with $BigInf_{it}$ and $NumSmallInf_{it}$. Figure 3 illustrates the process of constructing the instruments. Using $BigInf_{it}$ as an example, we first identify the big influencers who promoted product i between day t and day $t-k$, where $k = 0, 6, 13$. Next, before day $t-k$, we look for TikTok livestreams hosted by the focal big influencers to promote products in other categories. For example, if product i is a necklace, we look for livestreams for food, personal care products, household supplies, and so forth. Moreover, we set aside a two-week gap before day $t-k$ and focus only on livestreams that took place during the two-week period before the time gap. Finally, we use the total number of identified livestreams as an instrument for $BigInf_{it}$. To construct an instrument for $NumSmallInf_{it}$, we follow a similar procedure for small influencers. We instrument the interaction between $BigInf_{it}$ and $NumSmallInf_{it}$ with the interaction between their instruments (Wooldridge 2010).

We now discuss the validity of our instruments from two perspectives: instrument relevance and exclusion restrictions. Regarding instrument relevance, we argue that an influencer's sales performance and hiring cost are two of the most critical factors in determining whether a firm hires an influencer. If a large number of firms choose to hire an influencer at a particular point in time, the influencer is likely either performing exceptionally well in sales, charging an appealingly low price, or both, at that time. As a result, we anticipate that a firm will be more likely to collaborate with an influencer if that influencer is employed by many other firms in the recent past. To demonstrate the relevance of our instruments, we report the estimation results of the first-stage equations in Table W2 in the Web Appendix. The significant coefficient estimates indicate that our instruments have strong relevance. Additionally, the R^2 values of the first-stage equations increase by .048, .313,

and .279 after adding the instruments, indicating that the instruments are not weak (Rossi 2014).

To address the exclusion restrictions, we rely on two features of our instruments. First, when we construct the instruments, we leave a two-week gap before day t − k. This time gap ensures that the livestreams used for constructing our instruments and the livestreams that promote product i between day t and day t − k are unlikely to compete for viewers' attention. Therefore, the instruments should not impact product i's live sales except through $BigInf_{it}$ and $NumSmallInf_{it}$. Moreover, the instruments are constructed based on livestreams for products that fall into separate categories. For example, conditional on a variety of campaign characteristics, product fixed effects, and time fixed effects for which we have controlled, it is unlikely that live sales of shoes will be directly impacted by the number of livestreams that help firms promote food sales.

Empirical Findings

Overall Effects of Influencer Mix

Table 7 presents the estimation results of our main model, and each column corresponds to a time window for defining all variables. Overall, results in all columns suggest that both big influencers and small influencers have significantly positive effects on product sales. Based on the results in Column 1, employing any big influencer(s) increases live sales by 259.7% on average compared with observations that involve no big influencers. Employing an additional small influencer increases sales by 18.5% on average.[4] Therefore, it is not surprising to see that big influencers are sought by the vast majority of firms as long as their marketing budgets permit them to pursue this strategy.

Interestingly, the interaction between big and small influencers turns out to be significantly negative. According to results in Column 1 of Table 7, the sales effectiveness of an additional small influencer drops drastically, from 18.5% to 2.7%, when big influencers are also employed. Likewise, the marginal effectiveness of a big influencer gradually diminishes as more and more small influencers are employed to promote the same product. This negative interaction effect is consistent when the time windows for defining influencer strategies are longer, as shown in Columns 2 and 3. It indicates that big and small influencers will weaken each other's marginal positive sales effects when employed together in livestream commerce.[5] Consequently, even if a firm has an adequate marketing budget to pursue a combined influencer strategy, it should carefully evaluate the weaker but still positive sales effects as well as the relative costs of big and small influencers. We discuss this topic at greater length when we present our profit-budget analysis in the "Managerial Implications" section.

Order Effects of Influencer Mix

To gain deeper insights into the interaction between influencers, we examine the order effects of influencer mix at the product and session level. Specifically, we examine the effects of previous sessions hosted by big or small influencers on subsequent sessions hosted by other big or small influencers. This entails investigating the order effects of big influencers on big influencers, small influencers on big influencers, big influencers on small influencers, and small influencers on small influencers. To isolate the order effects on big influencers, we focus on livestream sessions hosted by big influencers on days when no small influencer promoted the same product. Similarly, for the order effects on small influencers, we focused on livestream sessions hosted by small influencers on days when no big influencer promoted the same product.

Our dependent variable is the log of product i's unit sales during one single livestream session j. The key variables of interest are $NumPrevSessionBig_{ij}$ and $NumPrevSessionSmall_{ij}$, representing the number of livestream sessions hosted by other big and small influencers prior to livestream session j for product i, respectively. Similar to our main analysis, we employ three time windows: previous livestream sessions within the past 1, 7, and 14 days. We create instruments for $NumPrevSessionBig_{ij}$ and $NumPrevSessionSmall_{ij}$ following Figure 3 and control for the same set of variables as in Equation 1.

The results in Table 8 show that the livestream sales generated by a big influencer are adversely affected regardless of whether the product had been promoted by big or small influencers previously. Within one day, the negative effects of previous big and small influencers appear similar, but the negative effects of previous small influencers diminish faster than those of previous big influencers. The effects of previous big influencers on small influencers are mostly nonsignificant. In contrast, when a small influencer followed another small influencer who had already promoted the product, there was initially a significantly negative effect on the later small influencer's sales. However, this effect becomes nonsignificant after one week. Overall, it is interesting that livestream sales generated by a big influencer are adversely affected by small influencers who promoted the same product previously, but not the other way around.

It is crucial to differentiate between the overall sales effects estimated in our main model and the order effects examined here. Consider two influencers, A and B, where influencer A hosted a livestream session prior to influencer B's session. The order effect measures the spillover effect of influencer A on influencer B's sales. In contrast, the overall effect captures a combination of direct effects (influencer A's and B's impact on their own sales) and spillover effects (A's impact on B's sales, and vice versa). While the order effects provide specific insights into the spillover between big and small influencers, the overall effects present a comprehensive understanding of the sales effectiveness of influencer strategies.

[4] In Column 1 of Table W8 in the Web Appendix, we use $NumBigInf_{it}$ instead of $BigInf_{it}$ and find that employing an additional big influencer ($b = .504$, $p < .01$) is far more effective than employing an additional small influencer ($b = .097$, $p < .01$) at increasing sales.

[5] In Column 2 of Table W8 in the Web Appendix, we show that the coefficient of $NumBigInf^2$ is significantly negative, whereas the coefficient of $NumSmallInf^2$ is nonsignificant. This suggests that big influencers will also weaken each other's marginal positive sales effects.

Table 7. Overall Effects of Influencer Mix.

	1-Day Window (1)	7-Day Window (2)	14-Day Window (3)
BigInf	2.597***	3.623***	4.094***
	(.166)	(.369)	(.609)
NumSmallInf	.185***	.092***	.077*
	(.030)	(.032)	(.042)
BigInf × NumSmallInf	−.158***	−.084**	−.073*
	(.032)	(.034)	(.043)
NumPdt	−.054***	−.078***	−.091***
	(.003)	(.007)	(.009)
CatFocus	.188***	.349***	.470***
	(.030)	(.075)	(.130)
Discount	.433***	.666***	.853***
	(.008)	(.018)	(.024)
Duration	.063***	.034***	.026***
	(.001)	(.005)	(.009)
$Time_1$	.404***	.884***	.954***
	(.015)	(.043)	(.072)
$Time_2$	.407***	.739***	.845***
	(.014)	(.041)	(.070)
$Time_3$	.578***	.791***	.834***
	(.012)	(.036)	(.060)
$Time_4$	.606***	.844***	.881***
	(.018)	(.056)	(.098)
NumVideo	.090***	.018***	.010***
	(.007)	(.002)	(.001)
VideoLike	.057***	.096***	.108***
	(.007)	(.012)	(.012)
Constant	2.049***	1.801***	1.388***
	(.067)	(.122)	(.189)
Product fixed effects	Yes	Yes	Yes
Holiday fixed effects	Yes	Yes	Yes
Day-of-week fixed effects	Yes	Yes	Yes
Month fixed effects	Yes	Yes	Yes
R^2	.0924	.0952	.0926
Number of observations	1,849,619	1,849,619	1,849,619
Number of products	108,001	108,001	108,001

*$p < .1$.
**$p < .05$.
***$p < .01$.
Notes: Robust standard errors are reported in parentheses.

Moderating Effects of Product and Campaign Characteristics

We estimate the moderating effects of two salient product attributes to which both firms and consumers pay attention: product category and price level. In Table 9 we report the sales effects of influencer strategies across the top four product categories observed in our data. We find that both big and small influencers are still influential in increasing product sales in all categories and are most effective in the food category. Moreover, the interaction effect between big and small influencers is negative and significant for most categories, but negative and nonsignificant for the clothing category. This finding suggests that clothing firms may safely hire big and small influencers together.

In Table 10 we report the sales effects of influencer strategies across four levels of live prices. We observe that more than half of the best-sellers in livestream commerce on TikTok are inexpensive products with prices below $1. There are, however, many high-priced products that sell very well through livestreams. Both big and small influencers are effective at selling products at all price levels, and the effect sizes increase as a product's price rises. Consumers are likely to depend more on influencers' endorsement when they purchase high-priced products as consumers need further persuasion to purchase those products.

Table 8. Order Effects of Influencer Mix.

	1-Day Window		7-Day Window		14-Day Window	
	Big Influencer (1)	Small Influencer (2)	Big Influencer (3)	Small Influencer (4)	Big Influencer (5)	Small Influencer (6)
NumPrevSessionBig	−.126***	.010	−.036***	.006*	−.024***	.003
	(.041)	(.020)	(.009)	(.004)	(.006)	(.002)
NumPrevSessionSmall	−.114***	−.007***	−.019***	−.001***	−.007**	−.000
	(.032)	(.003)	(.006)	(.000)	(.003)	(.000)
Control variables and fixed effects	Yes	Yes	Yes	Yes	Yes	Yes
Number of observations	483,317	2,212,461	483,317	2,212,461	483,317	2,212,461
Number of products	45,467	76,173	45,467	76,173	45,467	76,173

*$p < .1$.
**$p < .05$.
***$p < .01$.
Notes: Robust standard errors are reported in parentheses. Full estimation results are reported in Table W3 in the Web Appendix.

Table 9. Moderating Effects of Product Category.

	Clothing (1)	Beauty (2)	Food (3)	Household (4)
BigInf	1.934***	3.045***	4.846***	1.892***
	(.335)	(.395)	(.589)	(.181)
NumSmallInf	.117**	.362***	.485***	.142***
	(.059)	(.083)	(.135)	(.033)
BigInf × NumSmallInf	−.105	−.345***	−.465***	−.114***
	(.073)	(.089)	(.136)	(.035)
Control variables and fixed effects	Yes	Yes	Yes	Yes
Number of observations	627,019	224,703	411,772	210,868
Number of products	45,377	16,652	14,721	10,212

*$p < .1$.
**$p < .05$.
***$p < .01$.
Notes: Robust standard errors are reported in parentheses. Full estimation results are in Table W4 in the Web Appendix.

Table 10. Moderating Effects of Live Price.

	Below $5 (1)	$5 to $10 (2)	$10 to $100 (3)	Above $100 (4)
BigInf	2.313***	2.974***	4.268***	8.971***
	(.215)	(.222)	(.653)	(2.807)
NumSmallInf	.175***	.230***	.394***	3.147*
	(.035)	(.038)	(.151)	(1.865)
BigInf × NumSmallInf	−.153***	−.183***	−.364**	−4.015**
	(.038)	(.039)	(.159)	(2.020)
Control variables and fixed effects	Yes	Yes	Yes	Yes
Number of observations	878,981	422,694	541,409	6,535
Number of products	41,561	28,472	45,629	988

*$p < .1$.
**$p < .05$.
***$p < .01$.
Notes: Robust standard errors are reported in parentheses. Full estimation results are in Table W5 in the Web Appendix.

Table 11. Moderating Effects of Campaign Characteristics.

	(1)	(2)	(3)	(4)
BigInf	3.647***	2.204***	2.554***	3.221***
	(.277)	(.166)	(.172)	(.267)
NumSmallInf	.176***	.171***	.204***	.182***
	(.039)	(.028)	(.032)	(.038)
BigInf × NumSmallInf	−.214***	−.145***	−.171***	−.217***
	(.039)	(.030)	(.035)	(.038)
BigInf × NumPdt	−.109***			−.105***
	(.016)			(.016)
NumSmallInf × NumPdt	.010***			.010***
	(.001)			(.001)
BigInf × CatFocus		.930***		.865***
		(.243)		(.258)
NumSmallInf × CatFocus		.006		.006
		(.008)		(.009)
BigInf × Discount			.316***	.306***
			(.083)	(.084)
NumSmallInf × Discount			−.022***	−.024***
			(.003)	(.003)
NumPdt	−.036***	−.054***	−.054***	−.038***
	(.006)	(.003)	(.003)	(.006)
CatFocus	.192***	−.139*	.194***	−.107
	(.033)	(.085)	(.031)	(.089)
Discount	.422***	.427***	.397***	.390***
	(.008)	(.008)	(.025)	(.025)
Control variables and fixed effects	Yes	Yes	Yes	Yes
Number of observations	1,849,619	1,849,619	1,849,619	1,849,619
Number of products	108,001	108,001	108,001	108,001

*$p < .1$.
**$p < .05$.
***$p < .01$.
Notes: Robust standard errors are reported in parentheses. Full estimation results are in Table W6 in the Web Appendix.

Consistent with our main findings, the interaction effect between big and small influencers is significantly negative, and its magnitude also increases as the price level rises. Notably, for products priced above $100, the negative interaction effect exceeds the positive effect of hiring an additional small influencer, implying that firms that sell expensive products should not hire big and small influencers together when engaging in livestream commerce.

To estimate the moderating effects of campaign characteristics (NumPdt, CatFocus, and Discount), we include additional interaction terms between the moderators and the influencer variables in our main model. Importantly, we find that the moderating effects are very different for big and small influencers. Table 11 presents the results, and Table W7 in the Web Appendix summarizes the marginal effects of BigInf and NumSmallInf to aid in the interpretation of these moderating effects.

First, while the main effect of NumPdt is negative for both big and small influencers, the negative effect amplifies when any big influencer is hired and weakens when more small influencers are hired. As previously discussed, although promoting more products in a livestream may draw a larger audience, it is also likely to reduce consumers' trust in influencers. Moreover, the trust decline with regard to big influencers could be more severe because it is more difficult for big influencers to maintain their perceived trustworthiness as more firms collaborate with them.

Second, we find that the interaction with category focus is significantly positive for big influencers but nonsignificant for small influencers. We argue that consumers are more likely to perceive an influencer as an expert in a particular category if the influencer focuses solely on that category (high category focus). In contrast, an influencer is more likely to be viewed as highly commercially motivated if the influencer promotes a wide range of product categories (low category focus). Furthermore, unlike small influencers who can keep an intimate relationship with their followers, big influencers must rely on their own aura of expertise and trustworthiness to persuade their followers. Therefore, anything that hurts a big influencer's reputation will hurt the influencer's credibility.

Finally, while a product's sales will increase significantly when its price discount deepens, we find that big influencers

will benefit more significantly from deeper price discounts. The positive effect of price discounts gradually diminishes, however, when more small influencers are employed. Consumers may be more likely to rely on their close ties with small influencers than on price discounts when making purchase decisions. In contrast, consumers who watch livestreams of big influencers take advantage of the deeper discounts since big influencers often have greater bargaining power than small influencers.

Robustness Checks

To ensure the validity of our findings, we conduct extensive robustness checks, and the results are reported in the Web Appendix. First, our findings are consistent when using alternative influencer variables, such as using the number of big influencers, including the squared numbers of big and small influencers, using the number of influencers and their average number of followers, and using the number of previous influencers (see Tables W8 and W9 in the Web Appendix).

Second, we further address the endogeneity issues using two methods. In Column 1 of Table W10 in the Web Appendix, we control for the fit between an influencer and a product based on the influencer's TikTok content label and the product's category. For example, if the content label is fashion and the product category is clothing, the dummy variable for fit will be 1; however, the fit will be 0 if the category is food and the content label is fashion. In Column 2, we use an additional instrument based on whether or not an influencer was affected by pandemic-related lockdowns on day t. This instrument serves as a cost shifter for firms' influencer strategy because of the inherent difficulties faced by the influencer teams when collaborating with a firm during a lockdown. However, the lockdown of an influencer is unlikely to impact consumer demand since consumers can be from all over the country and are unlikely to be affected by the same lockdown. Both analyses yield consistent results.

Third, we control for product-influencer fixed effects, which capture unobserved product and influencer characteristics. In Column 1 of Table W11 in the Web Appendix, we show that big influencers are more effective than small influencers. In Columns 2 and 3 of Table W11, we find similar order effects.

In addition, we replicate our results using alternative threshold values for defining big influencers (half a million, two million, and three million followers), alternative dependent variables (products' sales revenues in livestreams and products' total sales on TikTok, which include sales generated by organic traffic to TikTok stores), and alternative model specifications (excluding all control variables, excluding the variable of livestream duration, and controlling for daily fixed effects). The results are reported in Tables W12–W18 in the Web Appendix.

Finally, we test subsamples for robustness and obtain consistent findings, shown in Table W19 in the Web Appendix. Column 1 focuses on products promoted by both big and small influencers during our data period, and Column 2 focuses on observations with no price discount so that we can tease out the sales effects of influencer strategies from those of price discounts.

Underlying Mechanisms

Our findings suggest that big influencers are more effective at generating product sales than small influencers and there is a negative interaction effect when big and small influencers are employed together. In this section, we provide supportive empirical and experimental evidence for some of our theoretical arguments regarding these empirical findings.

Empirical Analysis: Audience Reach and Conversion Rate

We argued that big influencers are more adept at reaching large audiences while small influencers are more effective at driving conversions. To test these arguments, we analyze how being a big influencer affects the number of viewers (i.e., audience reach) and the ratio of live sales to viewers (i.e., conversion rate) in each livestream. We find that BigInf exerts a significantly positive effect on audience reach but a significantly negative effect on conversion rate, in line with our theoretical arguments (see Table W20 in the Web Appendix).

Empirical Analysis: Follower Overlap

Our arguments for the interaction effect between big and small influencers are based on a critical premise that there can be a significant audience overlap between influencers. To substantiate this premise, we diligently gathered data on follower overlap, comprising 3,865 pairs of 981 influencers within our data set. We create two variables to quantify overlap: the number of overlapped followers between two influencers and a dummy variable that takes a value of 1 if the number of overlapped followers surpasses the median value, and 0 otherwise. We incorporate each variable, along with its interaction with BigInf × NumSmallInf, in Equation 1 and report the estimates in Table W21 in the Web Appendix. Remarkably, our findings reveal that the negative interaction between big and small influencers amplifies as the level of follower overlap between them increases. These empirical findings serve as robust evidence in support of our theoretical argument.

Online Experiment: Consumers' Trust in Influencers

We conducted a preregistered online experiment to provide compelling evidence demonstrating that consumers' trust in influencers, particularly big influencers, and their purchase intentions decrease when they learn that other influencers are promoting the same product. A total of 404 participants from Credamo who frequently watch livestreams in China completed the experiment and received financial compensation for their participation. Participants were randomly assigned to one of four conditions in a 2 (a big vs. small fictional influencer promoting a product in a livestream) by 2 (no awareness vs. awareness of other influencers promoting the same product) between-participant design. The details of our experiment design and results are reported in the Web Appendix.

Regarding the big influencer, the t-test results indicate a statistically significant decrease in both purchase intention

Table 12. Cost of Hiring Influencers (Per Product and Livestream).

	Obs.	Mean	SD	Min	25th Percentile	50th Percentile	75th Percentile	Max
Big influencers	82	$2,260	3,648	$15	$462	$798	$1,538	$20,012
Small influencers	236	$182	379	$5	$77	$77	$154	$4,615

($M_{noaware} = 5.58$, $M_{aware} = 5.29$, $p = .029$) and perceived trustworthiness ($M_{noaware} = 5.67$, $M_{aware} = 5.40$, $p = .021$) when participants were aware that other influencers were promoting the same product, in comparison to when they were unaware of this fact. However, there is no significant decrease in the perceived expertise of the big influencer ($p = .067$). As for the small influencer, whether or not participants learned that other influencers promoting the same product appears to have no significant impact on their purchase intention, perceived trustworthiness, or perceived expertise ($p > .05$).

In addition, when participants were unaware that other influencers were promoting the same product, the purchase intention ($M_{big} = 5.58$, $M_{small} = 5.04$, $p = .000$), perceived trustworthiness ($M_{big} = 5.67$, $M_{small} = 5.43$, $p = .033$), and perceived expertise ($M_{big} = 5.20$, $M_{small} = 4.80$, $p = .008$) for the big influencer are significantly higher than those for the small influencer. Interestingly, when participants were aware of other influencers, there is no significant difference between the big and small influencers in these aspects ($p > .05$).

Finally, participants believed that the firm had a larger marketing budget when a big influencer rather than a small influencer promoted the product ($p < .05$). Regardless of whether participants were presented with a big or small influencer, they thought the firm had a substantial marketing budget when they were aware that other influencers were promoting the same product ($p < .05$). These findings imply that consumers associate different influencer strategies with a firm's marketing efforts.

Overall, the experimental findings align with our theoretical arguments that consumers express greater concerns regarding the trustworthiness of big influencers compared with small influencers. In particular, when consumers were aware of other influencers promoting the same product, their trust in big influencers significantly decreases. However, this decrease in trust does not occur for small influencers under the same circumstances.

Managerial Implications

Our findings offer insights into the sales effectiveness of big and small influencers and general guidelines for designing their mix. In addition, we highlight a potential pitfall that could reduce the effectiveness of livestream commerce and influencer marketing: while many marketers believe that combining big and small influencers can generate the highest returns, we show that combining the two types of influencers may weaken each other's effectiveness in livestream commerce and thus the combined strategy may not always maximize marketing return on investment. While a combined strategy may still be effective at maximizing sales, the additional sales may come at a higher marketing cost.

Under what circumstances should brands and firms pursue a big-influencer-only strategy, a small-influencer-only strategy, or a combined strategy? We expect the answers to depend on the relative sales effectiveness and hiring costs of big and small influencers. We manually collect the hiring costs of a subset of our influencers from Xingtu. The hiring cost of a TikTok influencer posted on Xingtu is for promoting one product in a single livestream, and we collected cost data for 82 big influencers and 236 small influencers as summarized in Table 12. The average cost of hiring big influencers is more than ten times higher than that of hiring small influencers. But, as expected, the cost of hiring either type of influencer varies greatly.

Based on the cost data, we conduct a profit-budget scenario analysis to approximate the optimal profit under the three influencer strategies. We first create five cost scenarios reflecting varying levels of influencer costs as observed in the market we analyze: (1) medium-cost influencers (i.e., using median costs); (2) high-cost influencers (using costs at the 75th percentile); (3) low-cost influencers (using costs at the 25th percentile); (4) high-cost big influencers and low-cost small influencers; and (5) low-cost big influencers and high-cost small influencers. Second, we use the coefficient estimates of $NumBigInf_{it}$, $NumSmallInf_{it}$, and their interaction term and squared terms in Columns 2 and 3 of Table W8 in the Web Appendix to approximate the percentage change in sales generated by hiring an additional influencer. Third, we list all possible numbers of big and small influencers under each influencer strategy, with a maximum of 9 big influencers and 50 small influencers based on our data. Finally, for each possible mix of influencers, we compute the incremental budget, revenue, and profit and plot the efficient frontier. That is, each point along the x-axis represents a fixed marketing budget, and the corresponding point on the curve represents the optimal mix of influencers that yields the highest profit for that particular budget. We calculate the incremental revenue by multiplying the percentage change in sales units by the number of influencers and their average sales revenue in our data.

Figure 4 depicts the increase in profits with an increase in budget across the three influencer strategies under different cost scenarios. In Figure 4, Panel A, when the costs of big and small influencers are at the medium level, the big-influencer-only strategy performs similarly to the combined strategy, while dominating the small-influencer-only strategy. However, the combined strategy is slightly more profitable than the big-influencer-only strategy when the incremental budget is high. Panels B and C of Figure 4 show similar patterns for high-cost and low-cost scenarios. These findings imply that small influencers may have charged too much given their sales effectiveness, making the small-influencer-only and combined strategies less profitable. The combined strategy can beat the big-influencer-only strategy only when the costs of

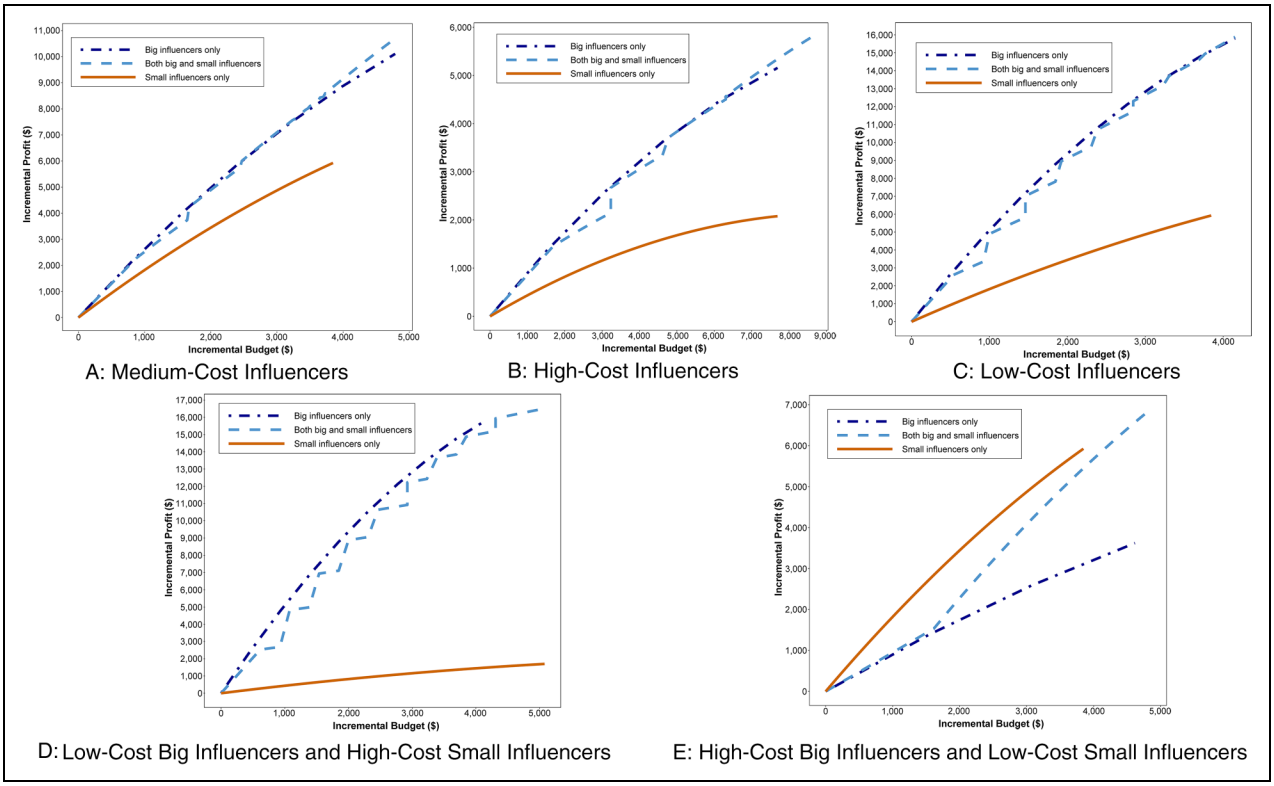


Figure 4. Optimal Profits Under Three Influencer Strategies.
Notes: The curves are right-truncated by the maximum number of influencers under each strategy.

hiring big influencers are high and the costs of hiring small influencers are low (see Figure 4, Panel E). However, the small-influencer-only strategy will be more profitable than the combined strategy in this scenario. Overall, the combined strategy does not have a clear advantage over the big-influencer-only and small-influencer-only strategies.

Armed with this analysis, a firm can determine the optimal strategy given a specific budget. If they know the costs of hiring big and small influencers, they can utilize the analysis in Figure 4 as the basis to identify the optimal strategy. While the big-influencer-only and small-influencer-only strategies are simple to implement, our findings on the order effects suggest additional guidelines when it comes to the combined strategy. Specifically, it is preferable to start with a big influencer to promote a product and then have small influencers follow them so that the firm can minimize the negative interaction effect. The firm should also select the influencers in such a way that the overlap in the influencers' follower bases is minimized to the greatest extent possible. Finally, even if the follower bases of multiple influencers do not overlap, platforms use algorithms to target consumers who may be interested in a promoted product (Zhao 2021). Audience awareness of a product being livestreamed by multiple influencers can undermine the impact of big influencers through decreased trust. As a result, a key recommendation for the firm when implementing a combined strategy is to not promote the livestreams indiscriminately to all users and followers of the hired influencers. Firms must be careful in how they promote the livestream campaigns.

There is, however, a note of caution in interpreting the scenario analysis. First, it should be noted that our understanding of influencer costs is not comprehensive, as these costs can include various sales incentives and negotiated expenses that may deviate from the figures posted on Xingtu. Second, we were only able to collect a subset of influencers' expenses, which may not accurately reflect the costs of all influencers. In addition, although the influencer cost is considered to be exogenous in our analysis, it is often determined through negotiations between the influencer and the firm and may also be contingent on the influencer's sales performance and other factors, such as price discounts. Finally, we do not observe the operational expenses associated with collaborating with different types of influencers, which may also significantly impact marketing efficiency (Hanssens and Pauwels 2016). Nevertheless, these factors may only lead to a shift in the cost designation of the influencers (a low-cost influencer may become a medium-cost influencer as a result of the additional expenses). Therefore, Figure 4 may still be utilized to determine the optimal strategy given the revised costs. Alternatively, a firm may choose a specific influencer strategy and then develop a budget plan to justify the strategy, which may then be utilized to select and negotiate with influencers.

In addition, our results regarding the moderating effects of product and campaign characteristics can provide specific recommendations for firms. For instance, when adopting a big-influencer-only strategy, firms should negotiate with big influencers to promote a smaller number of products from the same product category (except direct competitors) in a given

livestream. When using the small-influencer-only strategy, however, whether promoted products fall into the same category has little effect on product sales for small influencers. Firms should also be prepared to offer deep price discounts to consumers when collaborating with big influencers, as this is one of the main forces driving sales in big influencers' livestreams. Likewise, our insights should be valuable for influencers seeking to design the content of their marketing campaigns and increase product sales.

Finally, although our empirical context is TikTok in China, livestream commerce in other countries and on other platforms such as Facebook and Amazon are natural extensions because they share similar contextual elements with TikTok. Moreover, our findings can be qualitatively generalized to various forms of influencer marketing. Despite livestreaming's heightened interactivity and shopping-oriented nature, the fundamental facets of influencer marketing, such as the credibility and attractiveness of influencers, along with audience overlaps between influencers, are universal issues for all content forms. Overall, our findings provide important practical guidelines for both firms and influencers seeking to grow sales with influencer marketing.

Conclusions

The popularity of influencer marketing as well as livestream commerce creates not only opportunities but also challenges for marketers. An unresolved question is whether firms should collaborate with a single big influencer or multiple small influencers or even both. To answer this question, we obtain a unique and rich data set on TikTok livestreams, and our research yields several important findings on influencer mix. First, we demonstrate empirically that big influencers outperform small influencers significantly in terms of audience reach and product sales, whereas small influencers can generate higher conversion rates. Moreover, we find an interesting negative interaction effect between big and small influencers. While many marketers believe that big and small influencers may complement each other and generate increased sales, our results suggest that big and small influencers could weaken each other's sales performance when they promote the same product in livestreams. The mechanism leading to this negative interaction is consumers' decreased trust in big influencers as a result of consumers becoming aware of multiple influencers promoting the same product, due to either follower overlap or the platform's recommendation algorithm. Furthermore, we find that livestream sales generated by a big influencer are adversely affected by small influencers who promoted the same product previously, but not the other way around.

In addition, we show that the sales effectiveness and interaction effect of influencer mix vary across product categories and the effect sizes increase as product price increases. Also, we highlight that the number of products promoted in a livestream, the category focus of a livestream, and price discounts moderate the effectiveness of big and small influencers in different ways. Promoting a large number of products in a livestream reduces big influencers' sales of a focal product to a greater extent than it reduces small influencers' sales. Whereas big influencers generate higher sales of a given product when their livestreams focus exclusively on that product's category, small influencers' sales performance may not benefit from category focus. A deep price discount increases sales by big influencers more significantly than sales by small influencers.

This study is subject to several limitations that may provide opportunities for future research. First, we do not observe the exact costs that firms incur in employing influencers and using livestreaming platforms. However, marketing costs can vary widely among influencers with a complex structure, and should play a crucial role in firms' strategic decision making. Future research could consider how the cost structure involved in influencer marketing affects product sales and provide recommendations regarding optimal profits and influencer strategies. Second, in addition to the size of the follower base, there are other important influencer characteristics that may affect influencers' sales effectiveness, such as the quality of influencer content and the fit between the images of an influencer and a brand. However, we do not observe these influencer characteristics and thus control for them by using the instrumental approach as well as the product and influencer fixed effects. In the future, researchers can study the effects of influencer characteristics by collecting those data. Third, in essence, our estimates provide insights into the sales effectiveness of influencer strategies while operating within the confines of platforms' recommendation algorithms, a common environment for firms' marketing practices. However, the effectiveness of the influencer mix might change in the absence of these algorithms, which we leave to future research. In addition, our data focus on products among the top 2,000 daily best-sellers on TikTok for at least one day. In the future, scholars may provide more comprehensive findings on livestream commerce by including less popular products. Finally, we do not observe the prices or discounts of all products promoted in each livestream. Future studies may have better controls for product competition by gathering details of all products in each livestream.

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